A

Internship Report

On

Fabrication and Characterization of Force Sensor Using Shape Memory Polymer

Kyushu Institute of Technology, Japan

submitted in the partial fulfillment for the award of degree of

Bachelors of Technology

In

Computer Science and Engineering

By

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17005

Under the Supervision of

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Kyushu Institute of Technology, JAPAN

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Affiliated to DR. APJ ABDUL KALAM TECHNICAL UNIVERSITY LUCKNOW

DECLARATION

I hereby declare that this submission is my own work and that, to the best of my knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

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ABSTRACT

This project focuses on the fabrication and characterization of a variable sensitivity force sensor using Shape Memory Polymer (SMP). The sensor utilizes the property of SMP to change stiffness with temperature, enabling adaptive force sensing. A heating wire is used to control stiffness, while a strain gauge is used to measure force through resistance variation. This work highlights the importance of soft robotics in developing safe and adaptive systems for human interaction, particularly in healthcare applications.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to Dr. Kazuto Takashima for his guidance and support during my internship.

I am also thankful to the organization, Kyushu Institute of Technology, Japan, for providing me with the opportunity to work in a professional research environment and gain practical exposure to real-world applications of computer vision and machine learning.

I would like to extend my gratitude to my department and faculty members for their support and motivation during the course of this work. Finally, I am thankful to my family and friends for their continuous encouragement and support.

Introduction

Soft robotics is a branch of robotics that focuses on the development of robots using flexible and deformable materials instead of rigid components. These robots are inspired by biological systems and are capable of bending, stretching, and adapting to different environments.

One of the major applications of soft robotics is in human-robot interaction, particularly in healthcare and elderly care. In such applications, safety is a critical factor. Traditional rigid robots can cause harm due to their hard structure, whereas soft robots provide a safer alternative.

Force sensing is an important aspect of robotic systems, especially when interacting with humans. However, conventional rigid sensors are not suitable for soft robots as they restrict flexibility and fail under large deformations. Therefore, soft and adaptable force sensors are required.

This project focuses on the development of a variable sensitivity force sensor using Shape Memory Polymer, which can adjust its stiffness according to external conditions and provide accurate force measurement.

Sensor Design and Materials

The force sensor developed in this project is based on the use of Shape Memory Polymer (SMP), heating wire, and strain gauge.

Shape Memory Polymers are smart materials that can change their stiffness and shape when exposed to temperature changes. At lower temperatures, the material remains in a rigid state, while at higher temperatures it becomes soft and flexible. This property makes SMP suitable for variable sensitivity sensors.

A heating wire is embedded within the material to control its temperature. When current passes through the wire, heat is generated, which changes the stiffness of the SMP. This allows the sensor to adapt its sensitivity depending on the applied force.

A strain gauge is used to measure the deformation of the sensor. When force is applied, the material deforms, causing a change in the electrical resistance of the strain gauge. This change is measured and used to calculate the applied force.

The combination of these components enables the development of a sensor that can operate effectively across a wide range of forces.

Fabrication Process

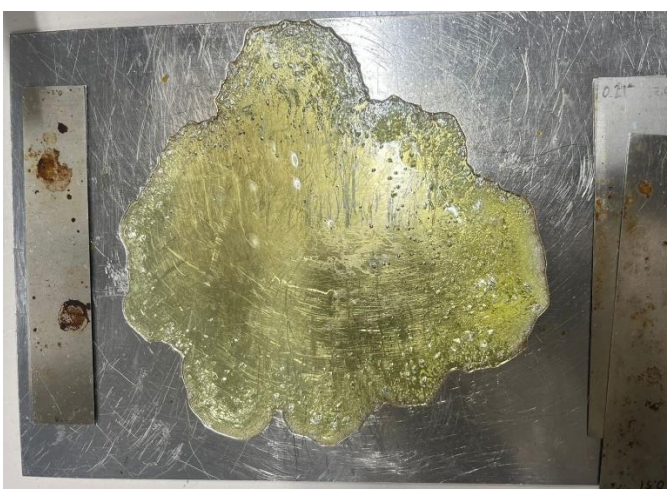
The fabrication of the force sensor involves several steps, including design, assembly, and integration of different components.

Initially, the sensor structure was designed using AutoCAD. This design helped in creating a proper layout for the sensor components. The heating wire layout was designed using Rhino 3D to ensure proper placement and uniform heating.

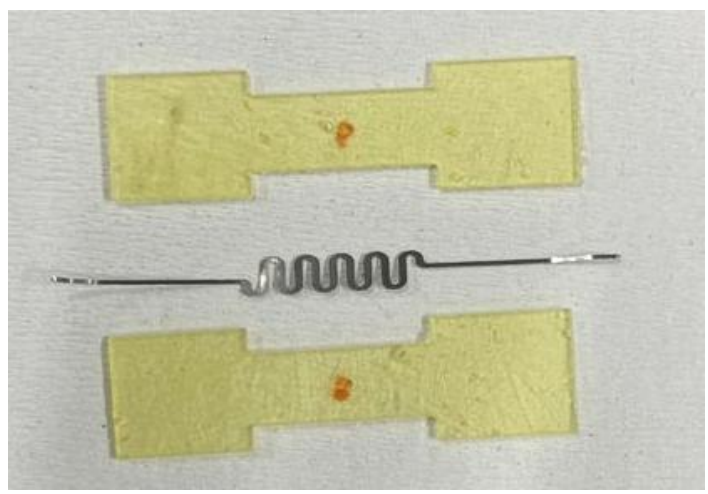
After the design phase, the heating wire was embedded within the Shape Memory Polymer material. Care was taken to ensure that the wire was properly aligned to achieve consistent heating.

The strain gauge was then attached to the sensor structure at a suitable position where maximum deformation occurs. Proper connections were made to measure resistance changes accurately.

Finally, the entire setup was assembled and tested to ensure proper functioning. The fabrication process required careful handling of materials and precise placement of components.



Compressed SMP Sheet



Cutting of SMP and heating wire

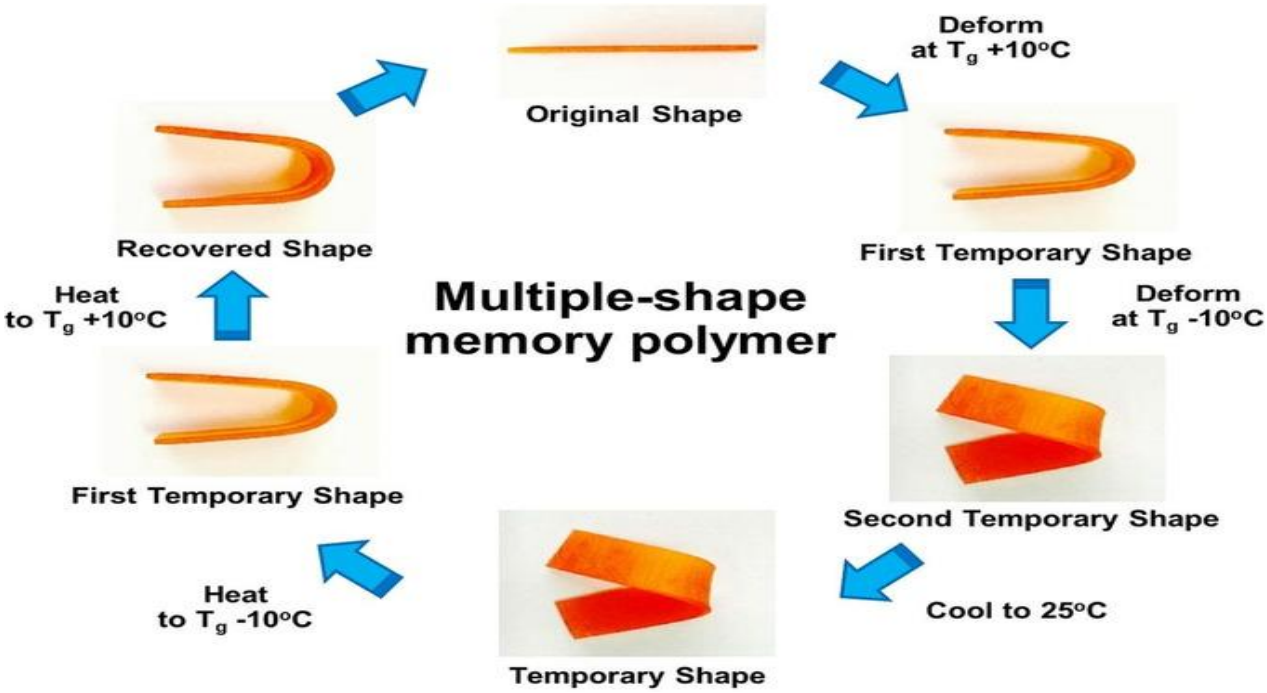
Working Principle

The working principle of the sensor is based on the relationship between force, deformation, and electrical resistance.

When no heat is applied, the Shape Memory Polymer remains in a rigid state, and the sensor has low sensitivity. When current is passed through the heating wire, the temperature increases, causing the SMP to soften. In this state, the sensor becomes more sensitive to applied force.

When an external force is applied, the sensor deforms. This deformation is detected by the strain gauge, which experiences a change in its electrical resistance. The resistance change is proportional to the applied force and can be measured using appropriate circuits.

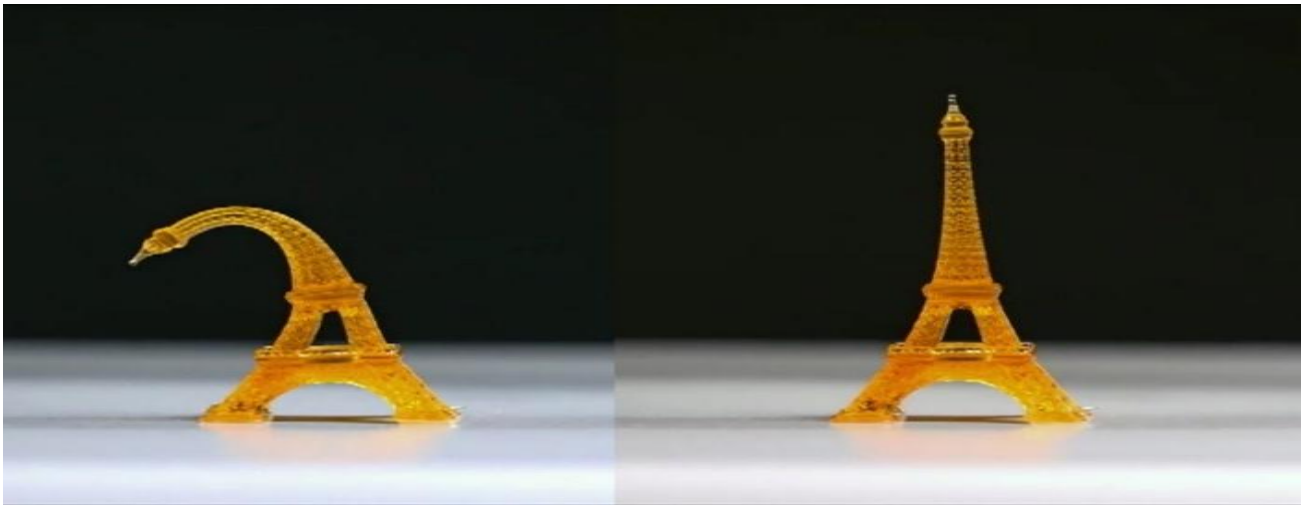
Thus, by controlling the temperature, the sensitivity of the sensor can be adjusted, making it suitable for different force ranges.

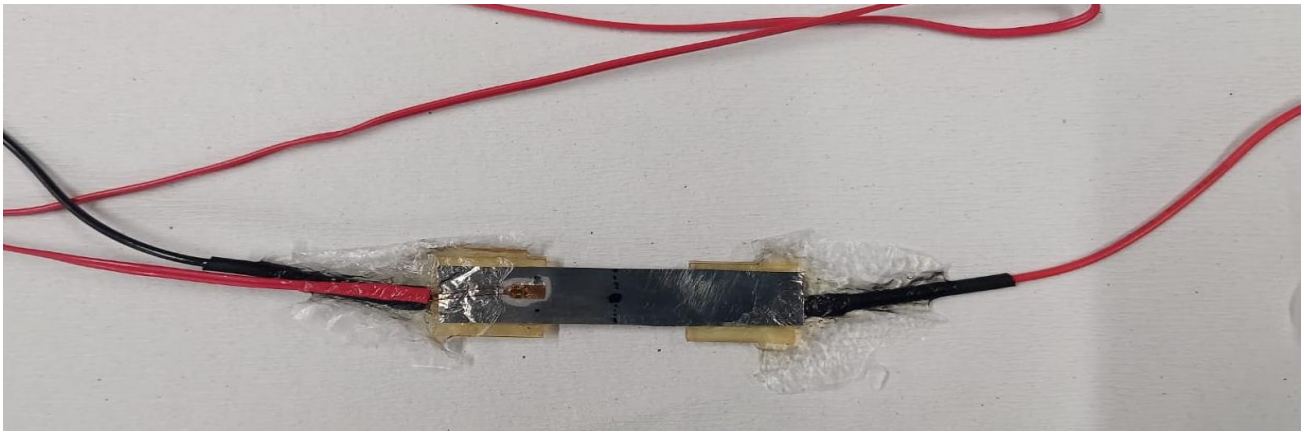


Results & Observations

During the testing phase, it was observed that the sensor exhibits different sensitivity levels depending on the temperature of the SMP.

At higher temperatures, the material becomes softer and shows higher sensitivity to small forces. At lower temperatures, the material is rigid and suitable for measuring larger forces.
Some challenges were observed in terms of non-linear response and calibration of the sensor. However, the overall performance demonstrates the effectiveness of using SMP for variable sensitivity force sensing.





Fabricated Force Sensor

Conclusion

The project successfully demonstrates the concept of a variable sensitivity force sensor using Shape Memory Polymer. The integration of heating wire and strain gauge allows the sensor to adapt its sensitivity based on external conditions.

This work highlights the importance of soft robotics in developing safe and adaptive systems for human interaction. It also provides valuable insight into the role of material properties in sensor design.

Future Scope

The performance of the sensor can be further improved by enhancing calibration techniques and improving material properties. Advanced sensing methods can be integrated to improve accuracy and reliability.

In the future, such sensors can be used in healthcare robotics, rehabilitation systems, and assistive devices. Integration with control systems and real-time monitoring can further enhance their applicability.

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Kyushu Institute of Technology

March 15, 2026

INTERNSHIP CERTIFICATE

This is to certify that **Mr. Somil Kathuria,** a student from Dronacharya Group of Institutions, India, has successfully and satisfactorily completed his internship in my laboratory during the training period of 2026-01-27 to 2026-03-15.

He worked on the research project entitled **"Fabrication and Characteri -zation of Variable-sensitivity Force Sensor Using Shape-memory Polymer"**.

During the internship, he demonstrated excellent skills and strong motivation for learning and acquiring new knowledge.

I wish him good luck and all the success in his life and future career.

Kazuto Takashima

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